

Hierarchical Echinus-like Cu-MFI Catalysts for Ethanol Dehydrogenation

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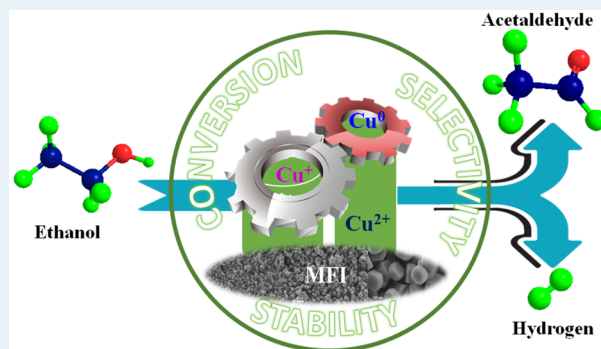
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ABSTRACT: Copper-based catalysts have been widely considered in the non-oxidative dehydrogenation of ethanol but suffer from deactivation troubles. Herein, a hierarchical echinus-like Cu-MFI-AE catalyst was developed, which exhibited a high yield of acetaldehyde and unprecedented stability. The formation of Cu–O–Si on the interface of Cu and hierarchical MFI avoids the aggregation of Cu nanoparticles and stabilizes the Cu⁺ species. The Cu⁺ and Cu⁰ species perform a synergetic mechanism, with Cu⁺ involving into acetaldehyde generation and Cu⁰ serving as H₂ liberation sites. The utilization of interfacial interactions is an effective method in stabilizing metal nanoparticles without affecting the active sites for ethanol conversion.



KEYWORDS: Cu-MFI-AE, ethanol, non-oxidative dehydrogenation, stability, interfacial interaction

Acetaldehyde is an important platform intermediate to produce a variety of value-added chemicals. Typically, the production of acetaldehyde is realized by the Wacker process via the oxidation of ethylene originated from fossil resources.^{1–3} Currently, the rapid development of “green” ethanol from biomass makes the direct conversion of ethanol to acetaldehyde an elegant method. Meanwhile, the elementary step of ethanol dehydrogenation to acetaldehyde is also inevitable in the upgrading of ethanol to longer carbon chain products, including *n*-butanol, 1,3-butadiene, and hydrocarbons.^{4–9}

Copper catalysts have shown great potential in oxidative dehydrogenation of ethanol to acetaldehyde because of the high activity of Cu species in O–H and C–H bonds oxidative cleavage.^{10–13} Specifically, the surface Cu⁺ species are involved in activating the molecular O₂ to the peroxide-type sites, which participate in catalyzing the rate-determining step of C–H bond cleavage. As such, over a Au/MgCuCr₂O₄ catalyst, the yield of acetaldehyde was reported to be ~95% at 523 K in 500 h time on stream (TOS).¹⁴ Although many progresses have been made in oxidative dehydrogenation, the non-oxidative dehydrogenation of ethanol to acetaldehyde is more attractive in view of the atom economy and viability of the reaction. Over Cu catalysts on the supports of rice husk ash,^{15,16} silica,¹⁷ mesoporous carbon, N-containing carbon- and silica carbon complex,^{18,19} > 90% acetaldehyde selectivity was obtained. Nevertheless, owing to the low Tamman temperature of Cu, most Cu particles are prone to be deactivated because of the aggregation of Cu in the presence of H₂ byproduct at elevated

temperatures.^{20,21} Moreover, various Cu species, including Cu⁰, Cu⁺, and Cu²⁺, are observed in Cu-based catalysts and are indispensable during the dehydrogenation reaction. The instability of Cu species in the presence of produced H₂ also leads to the challenge in identifying the role of Cu species in ethanol conversion. Hence, it is highly desirable to develop an efficient catalyst with great stability in the non-oxidative dehydrogenation of ethanol to acetaldehyde.

In this research, a hierarchical echinus-like catalyst of Cu-MFI-AE (silicalite-1) was developed for the non-oxidative dehydrogenation of ethanol to acetaldehyde. With this newly developed catalyst, the selectivity to acetaldehyde reached 93% under a conversion of 97% at 523 K and a WHSV (weight hourly space velocity) of 0.53 h^{–1} in the 500 h TOS. This unexpected activity of Cu-MFI-AE was attributed to the synergistic effect between Cu⁺ and Cu⁰ species on Cu-MFI-AE catalysts, which activated the O–H and the nearby C–H bonds, and promoted the H atoms migration and combination, respectively. Meanwhile, the strong interfacial interactions between Cu and crystallized MFI via Cu–O–Si bonds formation endowed the catalyst with an unprecedented stability.

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Figure 1A depicts the SEM image of the MFI support, in which a hexagonal platelet morphology was observed with two

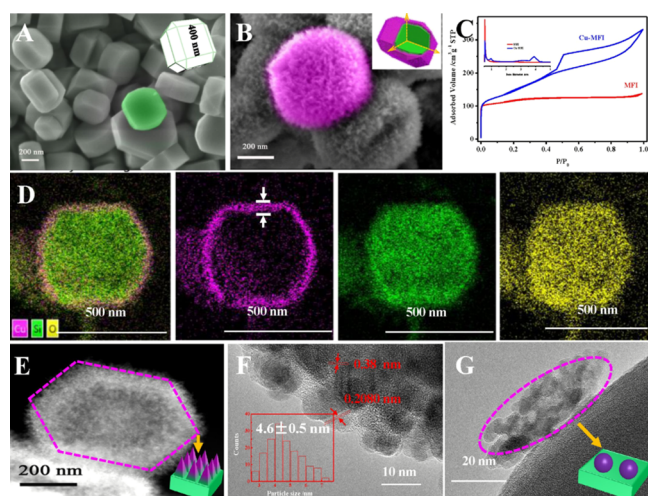


Figure 1. Physical texture of MFI support, Cu-MFI-AE and Cu/MFI-IM catalysts. SEM image of (A) MFI, (B) Cu-MFI-AE and their schematic illustration (inset); (C) full nitrogen sorption isotherms of the MFI and the Cu-MFI-AE catalyst with their pore size distribution (inset); (D) EDS elemental mapping of Cu, Si, and O for the Cu-MFI-AE film; (E) HAADF image of the Cu-MFI-AE catalyst and their schematic illustration of the thorns (inset); (F) HRTEM image of the Cu-MFI-AE catalyst, the size distribution of the Cu particles, and typical Cu species (inset); (G) HRTEM image of the Cu/MFI-IM catalyst and their schematic illustration (inset).

distinct (010) and (100) faces.^{22,23} Additionally, the MFI zeolite was highly crystallized (Figure S1) with smooth surfaces. After the introduction of Cu species by the ammonia evaporation (AE) method, the Cu-MFI-AE catalyst exhibited an echinus-like morphology, as shown in Figure 1B. According to the diameter difference between MFI (~400 nm) and Cu-MFI-AE (~500 nm), the enlarged thorn of the echinus-like catalyst was estimated to be ca. 50 nm (Figure S2). The introduction of Cu also changed the pore structure of MFI (Figure 1C, Table S1) because of the desilication process under basic conditions.^{24,25} In addition to the typical micropore structure of MFI, a distinct hysteresis loop was also observed in the isotherm range of ca. 0.4–1.0 P/P_0 , indicating the presence of hierarchical porosity (inset) with both mesopores and micropores.

To disclose the distribution of Cu species, the Cu-MFI-AE catalyst was dispersed in resin and cut into thin films (50 nm) for the EDS mapping. As shown in Figure 1D, the Cu species mainly located on the outlayer of the MFI zeolite, constructing a ca. 80 nm circle of Cu species. While the silica and oxygen elements were uniformly distributed throughout the Cu-MFI-AE catalyst. The echinus-like and hexagonal platelet structure of the Cu-MFI-AE catalyst was also confirmed by the HAADF image (Figure 1E), and the Cu particles on the thorn were highly dispersed with an average particle size of 4.6 nm (Figure 1F). Apart from the typical Cu (111) lattice spacing of 0.2080 nm (JCPDS, PDF no. 00-001-1241), a larger lattice spacing of >0.38 nm was also observed, which was attributed to the non-stoichiometric Cu–O–Si species. The catalyst was further characterized by aberration-corrected HAADF-STEM, which showed clearly the structure of Cu and Cu–O–Si species (Figure S3). Meanwhile, such a formation of Cu–O–Si was

also confirmed by the ^{29}Si MAS NMR, where a chemical shift of Si was observed when surface Si–OH were replaced by Si–O–Cu entities (Figure S4). In contrast, over the Cu/MFI-IM catalyst synthesized by the incipient-wetness impregnation (IM) method, the Cu particles aggregated on the smooth surface of MFI with a particle size as large as 40 nm (Figure 1G) because of their relatively weak interactions with the MFI support. Accordingly, owing to the strong Cu–O–Si interfacial interactions between Cu nanoparticles and MFI, the Cu-MFI-AE catalyst possesses a hierarchical echinus-like structure with a better dispersion of Cu species, which would endow it with unusual performance in ethanol conversion.

Dehydrogenation of ethanol to acetaldehyde was then evaluated on these Cu-based catalysts. As shown in Figure 2A, over the Cu-MFI-AE catalyst, the ethanol conversion and

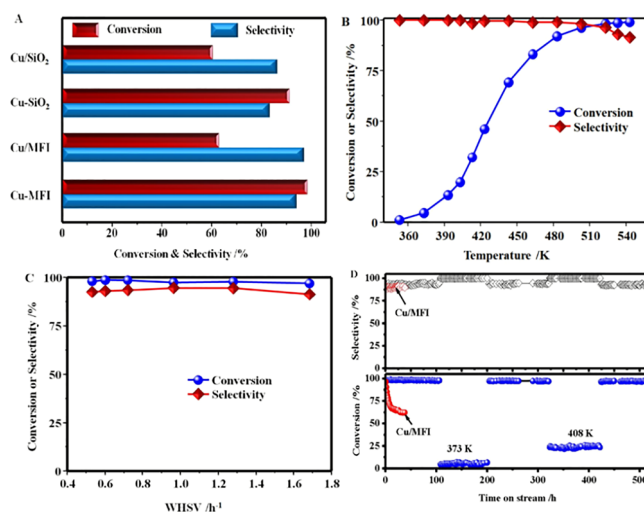


Figure 2. Dehydrogenation of ethanol over the Cu based catalysts. (A) Performance of the Cu based catalysts (523 K, WHSV = 0.53 h^{-1} , after 24 h reaction); (B) Effect of the reaction temperature (Cu-MFI-AE catalyst, 423–543 K, WHSV = 0.53 h^{-1}) on ethanol conversion and acetaldehyde selectivity; (C) Effect of the WHSV on ethanol conversion and acetaldehyde selectivity (Cu-MFI-AE catalyst, 523 K); (D) Ethanol conversion and acetaldehyde selectivity vs TOS over Cu-MFI-AE and Cu/MFI-IM catalysts (523 K, WHSV = 0.53 h^{-1} , trace byproducts include ethylene, butyraldehyde, and ethyl acetate).

acetaldehyde selectivity reached 97.9% and 93.1%, respectively (with hydrogen as the coproduct, Figure S5), close to the theoretical equilibrium conversion (Supporting Information 1.3). In contrast, the ethanol conversion was only 64.4% with a lower acetaldehyde selectivity (89.4%) over the Cu/MFI-IM catalyst under the same reaction conditions. Moreover, the Cu-MFI-AE catalyst demonstrated a lower apparent activation energy than the Cu/MFI-IM counterpart (Figure S6). The discrepancy of catalytic performance of these two catalysts should be attributed to the different catalyst structures caused by the distinctive preparation methods.²⁶ To confirm this hypothesis, SiO_2 was selected to load Cu species by AE and IM methods for ethanol conversion. The AE method prepared Cu- SiO_2 -AE catalyst demonstrated a higher activity than the IM counterpart (ethanol conversion: 91.1% vs 60.1% on Cu- SiO_2 -AE and Cu/ SiO_2 -IM, respectively). Figure 2B showed the temperature dependence of ethanol conversion and acetaldehyde selectivity over the Cu-MFI-AE catalyst. The ethanol conversion gradually increased with increasing reaction temperatures and exceeded 91% at 483 K. The acetaldehyde

selectivity slightly decreased but still surpassed 91% in the temperature range of 423–543 K. The WHSV did not significantly change the ethanol conversion and acetaldehyde selectivity in the range of 0.53–1.68 h⁻¹, still reaching >96% ethanol conversion and >91% acetaldehyde selectivity at 523 K (Figure 2C). Additionally, the catalyst prepared by the AE method had a superior stability to that of the IM method catalyst. In Figure 2D, the reaction was conducted at different temperatures with diverse ethanol conversions to detect the stability of the Cu-MFI-AE catalyst. In the first 100 h TOS at 523 K, the acetaldehyde selectivity leveled off around 93% at the ethanol conversion of 97%. Then, the ethanol conversions changed to ca. 5% and 22% at 373 and 408 K in the second and fourth 100 h TOS. After elevating the temperature again to 523 K, the ethanol conversion and acetaldehyde selectivity returned back to the initial values, demonstrating the unprecedentedly high stability of the Cu-MFI-AE catalyst in the 500 h TOS (Figure S7). In contrast, the rapid deactivation of the Cu/MFI-IM catalyst was observed in the first 38 h TOS.

The used Cu-MFI-AE catalyst was characterized by XRD and HAADF-STEM. It possessed the same crystallinity and morphology to these of the fresh catalyst (Figure S1 and S8A,B). More importantly, the small size of Cu species remained (Figures S8C,D), indicating the excellent stability of Cu-MFI-AE in ethanol dehydrogenation. For the Cu/MFI-IM catalyst, the irreversible aggregation of Cu nanoparticles was identified (Figures S9–S11), accounting for the rapid decrease in ethanol conversion.

From our results, the AE method plays a pivotal role in anchoring Cu species for the high stability of catalysts in ethanol conversion. The AE method was conducted in a high pH solution, which deprotonated the surface –OH of silica to adsorb the Cu species, forming the (NH₃)–Cu–O–Si precursor. After the calcination process, strong interfacial interactions between Cu species and the MFI support were constructed via the Cu–O–Si bonds (Supporting Information 1.3; Figure S3, ²⁹Si MAS NMR spectra).^{27,28} The strong interfacial interaction between Cu and Si via the Cu–O–Si bonds inhibited the Cu nanoparticles aggregation.^{29,30} For the MFI support, it kept a high crystallization and hierarchical structure after the AE process, not only providing a rather stable silica matrix to interact with Cu particles to lessen the deactivation of catalysts^{31–33} but also minimizing the steric and/or diffusional hindrance to facilitate the product transfer.^{34,35} Additionally, the acidity of the MFI support was rather weak, preventing the side reaction of ethanol dehydration.³⁶ As a result, the uniformly distributed Cu particles, the hierarchical porosity, and the strong interaction between Cu species and MFI support enabled this Cu-MFI-AE catalyst behaves as a best candidate in ethanol non-oxidative dehydrogenation in different SiO₂ supported catalysts (Figures S12–14).

To get insight into the active sites of the Cu-MFI-AE catalyst in ethanol non-oxidative dehydrogenation, the chemical states of Cu species were investigated with Cu K-edge XANES spectroscopy. As shown in Figure 3A, besides the Cu²⁺ and Cu⁰ species, a distinctive peak at 8981.5 eV attributed to Cu⁺ species (with the first derivative XANES peak at 8980.5 eV in Figure 3B) was observed over the Cu-MFI-AE catalyst. By fitting the XANES spectra at the K-edge of Cu into different chemical states of copper, the amount of Cu⁺ was estimated to be 35.9% (Table S2 and Figure S15). The presence of Cu⁺ was further reflected by H₂-TPR (Figure S16), in which a lower reduction peak was observed over the

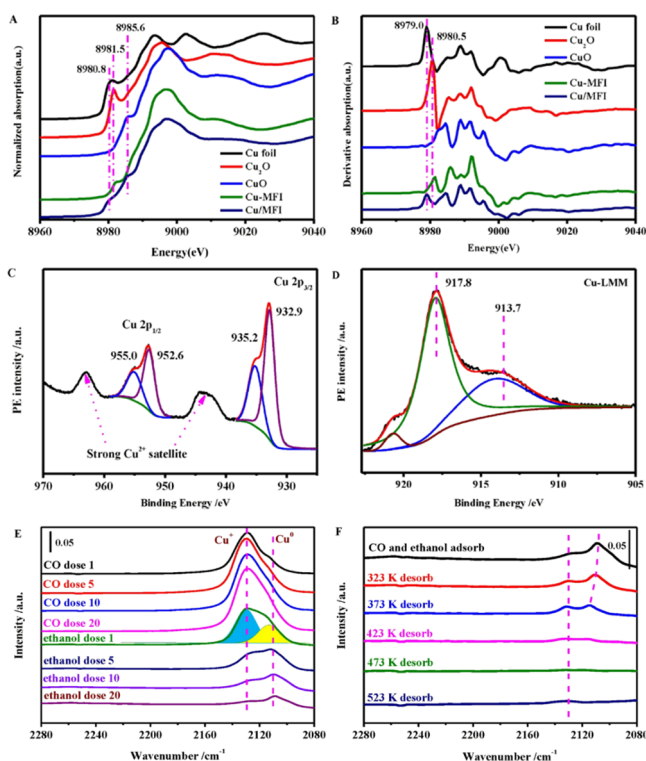


Figure 3. Chemical structure of Cu based catalysts. (A,B) Normalized and First derivative XANES spectra at the K-edge of the Cu foil, Cu₂O, CuO, Cu-MFI-AE, and Cu/MFI-IM catalysts; (C) Cu 2p XPS spectrum of the used Cu-MFI-AE catalyst; (D) Cu LMM XAES spectrum of the used Cu-MFI-AE catalyst; (E) DRIFTS analysis of sequential adsorbed CO and ethanol over the Cu-MFI-AE catalyst; (F) DRIFTS analysis of the desorption process at different temperatures over the Cu-MFI-AE catalyst.

Cu-MFI-AE and Cu-SiO₂-AE samples than those prepared by the IM method.^{37,38} Even for the Cu²⁺ species of Cu-MFI-AE (62.8%), it also behaves differently from the benchmark of CuO, arising from the Cu–O–Si bonds in Cu-MFI-AE rather than the Cu–O–Cu bonds in CuO. These can be the other evidence of the strong interfacial interactions between metallic Cu and MFI support. Two obvious separated peaks at 8980.8 and 8985.6 eV were detected over the Cu/MFI-IM catalyst (Figure 3A,B), which should be attributed to the Cu⁰ and Cu²⁺ species, respectively. The Cu⁰ and Cu²⁺ connected with –O–Cu are the dominated species (Cu⁰: 30%, Cu²⁺: 65.9%), with the content of Cu⁺ being negligible (4.2%) in the Cu/MFI-IM catalyst.

In view of the possible reduction Cu species during the H₂-abundant dehydrogenation condition, the chemical state of superficial Cu species on Cu-MFI-AE was further investigated by the *operando* X-ray photoelectron spectroscopy (XPS) (Figure 3C). Two peaks at 935.2 and 932.9 eV were observed, which should be attributed to the Cu²⁺ (35%), and Cu⁰ and/or Cu⁺ (65%) species, respectively. As evidenced by the overlapping peaks at 913.7 and 917.8 eV with the Auger Cu LMM spectrum (Figure 3D), the coexistence of Cu⁺ and Cu⁰ with a Cu⁺/Cu⁰ ratio of 0.47 was confirmed on the Cu-MFI-AE sample under *operando* conditions.³⁹ Accordingly, the Cu⁰, Cu⁺, and Cu²⁺ species are concomitant on the surface of Cu-MFI-AE sample under reaction conditions.^{40,41}

The anchoring sites of reactant on Cu sites were explored by DRIFTS using CO as a probe molecule. As shown in Figure

3E, two distinct bands were observed at 2129 and 2111 cm^{-1} , which were assigned to the CO adsorption on Cu^+ and Cu^0 sites, respectively. Further introducing ethanol caused a great decrease in the amount of CO adsorbed on Cu^+ sites, with gaseous CO emission as detected by the *on line* MS (Figure S17). In contrast, the band of CO adsorption on Cu^0 sites had little change. Thus, Cu^+ species are responsible for the ethanol adsorbing and activating in this reaction. Based on the interactions between Cu^+ and O containing species, the ethanol was partially activated to form the Cu-ethoxyl species. The as-formed Cu-ethoxyl species were rather stable at ≤ 523 K and gradually converted at higher temperatures (Figure 3F and Figure S18). Correspondingly, ethoxyl species were detected at the wavenumbers of 2956 and 2857 cm^{-1} , confirming the key role of $\text{Cu}(+)$ -ethoxyl species in this reaction.⁴² Additionally, the bands of CO on Cu^0 species gradually shifted to higher numbers with desorption temperature increasing and disappeared at 423 K. The blue shift of the Cu-CO bands indicates the weakened interaction between Cu^0 and CO, which might be attributed to the coadsorption of H atoms derived from the nearby C–H bonds at higher temperatures. Similar to the synergistic effect of $\text{Cu}^+/\text{Cu}^{x+}$ and Cu^0 species in CO_2 and ethanol conversion,^{43,44} over the Cu-MFI-AE catalyst, the Cu^+ adsorbed and activated the O–H group of ethanol, and the Cu^+/Cu^0 promoted the C–H bond cleavage and H atoms transfer, forming a catalytic cycle for the conversion of ethanol to acetaldehyde and hydrogen.

To gain an atomic level understanding of the ethanol dehydrogenation mechanism over the Cu-MFI-AE catalyst, the elementary steps for ethanol dehydrogenation on Cu (111) and Cu_2O (100) surfaces were explored theoretically. As shown in Figure 4A, over the Cu (111) surface, the barriers for O–H and C–H bonds cleavage are 1.06 and 1.36 eV, respectively, consistent with the previous report and confirming the rate-determining step of C–H bond cleavage in ethanol dehydrogenation.¹⁷ In contrast, the barriers for O–H and C–H bonds cleavage over the Cu_2O (100) surface are only 0.19 and 0.83 eV, respectively, indicating the preferable adsorption and activation of ethanol over the Cu^+ sites, in agreement with the experimental observation. Meanwhile, the combination of H adatoms to release gaseous H_2 is crucial to the final product and the completeness of catalytic cycle over the catalyst. An energy barrier on Cu_2O (100) surface is as high as 1.99 eV. While, it is only 0.98 eV over the Cu (111) surface, much lower than that on the Cu_2O . Therefore, it can be supposed that the H atoms undergo migration to metallic Cu sites and combine to H_2 product therein. As a result, the coexistence of Cu^+ and Cu^0 on the catalyst may account for the high activity of Cu-MFI-AE catalyst. The rapid migration of H from Cu^+ to Cu^0 also avoids the reduction of Cu^+ to metallic Cu and enhances the unprecedented stability in ethanol dehydrogenation. Otherwise, the Cu^+ species were rapidly reduced to metallic Cu, and caused the particle aggregation, as previously reported.⁴⁵ Based on the catalyst structure analysis and theoretical calculations, a possible reaction pathway for the dehydrogenation of ethanol was proposed in Figure 4B. Ethanol was preferably adsorbed on the Cu^+ sites of catalysts, which weakened the O–H bond and facilitated its cleavage. Afterward, the absorbed ethanol was transformed to $\text{Cu}(+)$ -ethoxy and H atom, which were adsorbed on Cu^+ and Cu^0 sites, respectively. The C–H bond of $\text{Cu}(+)$ -ethoxy was then activated by Cu^+ to produce the adsorbed acetaldehyde, and H atom was transferred to Cu^0 for combination.⁴⁶ Finally, the

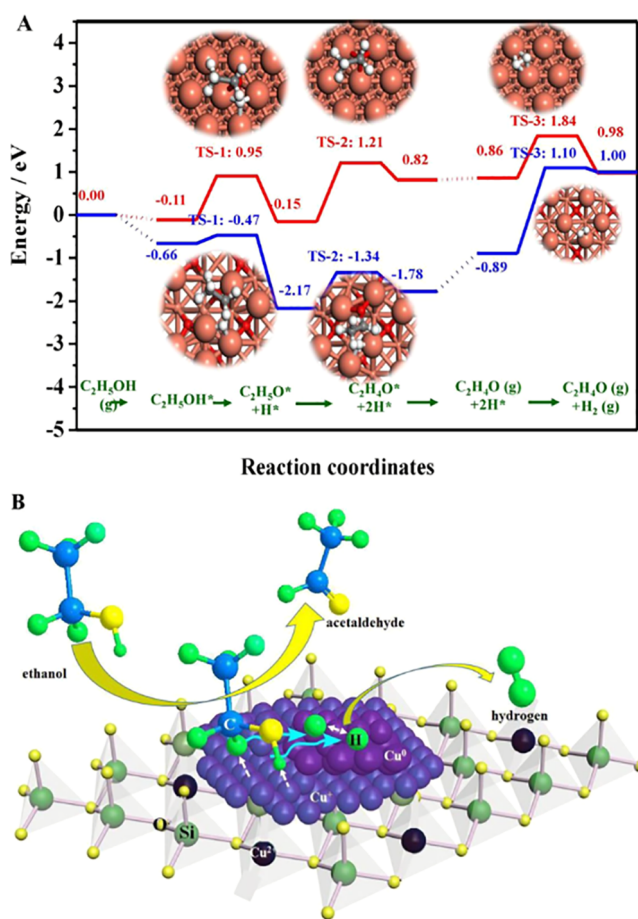


Figure 4. Profiles for dehydrogenation of ethanol to acetaldehyde. (A) Energy profile for ethanol dehydrogenation on Cu (111) (red) and Cu_2O (100) (blue) and (B) A schematic illustration of ethanol dehydrogenation on the Cu-MFI-AE catalyst (interface). All energies are relative to an ethanol molecule in the gas phase and a clean Cu or Cu_2O surface. Intermediates adsorbed on surface are marked with an asterisk.

adsorbed hydrogen on Cu^0 and the acetaldehyde on Cu^+ were desorbed to the gas phase. Different from Cu–Cr based catalysts,⁴⁷ synergistic effect of Cu^+ and Cu^0 species over the Cu-MFI-AE catalyst ensured the ethanol dehydrogenation reaction, rather than the esterification reaction.

In conclusion, a hierarchical echinus-like Cu-MFI-AE catalyst was synthesized for the non-oxidative dehydrogenation of ethanol. Acetaldehyde was obtained at 93% selectivity with 97% ethanol conversion in the 500 h TOS at 523 K and a WHSV of 0.53 h^{-1} . The Cu-MFI-AE catalyst possesses the capability of balanced Cu^+ and Cu^0 species, which activated the O–H, C–H bonds of the adsorbed ethanol, and promoted the H atoms transfer and combination, respectively. The strong interfacial interactions was constructed between Cu species and the crystallized MFI support via the Cu–O–Si bonds in the hierarchical structure, which prevented the reduction of Cu^+ , inhibited the Cu^0 aggregation, facilitated the product transfer, and resulted in the high stability of the Cu-MFI-AE catalyst.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.0c03860>.

Experimental procedures, Supporting Figures (S1–S18), Tables (S1–S2), and references (PDF)

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Author Contributions

J.P., M.Z., and T.Z. conceived and designed the project. J.P. performed the experiments. X.Y. carried out the theoretical calculations. H.Y. and X.L. performed the X-ray absorption experiments. J.S. and Y.W. contributed to the FT-IR characterization. J.P. and C.W. contributed to the structure characterizations. J.P., M.Z., X.Y., and T.Z. analyzed the experimental data and prepared the manuscript. All authors reviewed and contributed to the manuscript.

Notes

The authors declare no competing financial interest.

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